New York City Yellow Taxi Data

Exploratory Data Analysis Report

Optimising NYC Taxi Operations (2023)

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Assignment: EDA NYC Taxi Analysis

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Problem Statement: -

The objective of this project is to analyse New York City taxi trip data to identify patterns in demand, revenue generation, passenger behaviour, and operational inefficiencies.

By analysing factors such as pickup time, location, trip distance, passenger counts, and pricing behaviour, the study aims to understand how taxi services operate across different times and locations in the city.

The insights derived from this analysis can help taxi operators optimise vehicle routing, improve dispatch efficiency, and develop effective pricing strategies to better match supply with customer demand.

Assumptions Made: -

1. Data was sampled at 1% per hour per day using stratified sampling to keep it manageable while maintaining fair representation across all time periods.

2. Trips with zero fare amount and zero distance were removed as invalid.

3. Passenger count of 0 was treated as invalid and removed.

4. RatecodeID = 99 was removed as it is not defined in the data dictionary.

5. Extremely large trip distances or fare values were treated as outliers and excluded from analysis.

6. Passenger counts greater than the expected maximum value were assumed to be data entry errors and removed.

7. Two airport fee columns found in the data were merged into one.

8. Negative values in monetary columns were removed as data entry errors.

9. Only 2023 data was used.

Analysis Approach: -

The analysis follows a structured EDA pipeline covering the following steps:

1. Data Loading: Loaded 12 monthly parquet files containing approximately 36 million rows total and applied stratified sampling to produce a representative dataset.

2. Data Cleaning: Fixed column issues, handled missing values, removed outliers and negative values to ensure data quality.

3. Feature Engineering: Created new columns including pickup_hour, day_of_week, month, quarter, trip_duration, fare_per_mile, tip_percent, route, distance_tier and day_type to enable deeper analysis.

4. General EDA: Analysed temporal patterns (hourly, daily, monthly), financial metrics and geographical distribution using choropleth maps.

5. Detailed EDA: Deep-dived into operational efficiency, pricing strategy and customer experience factors.

6. Conclusions: Derived actionable business recommendations for routing, cab positioning and pricing strategy.

Data Cleaning and Preparation: -

Before performing analysis, the dataset was cleaned and processed to ensure accuracy and reliability. Raw data contained several quality issues that were identified and fixed before analysis

• Sampling the dataset using hourly stratified sampling to reduce size while preserving demand distribution as the each of the 12 monthly parquet files have approximately 3 million rows making a total of around 36 million rows.

• Removing trips with zero or negative fare values.

• Removing unrealistic trip distances and extreme outliers.

• Handling invalid passenger counts and correcting inconsistent entries.

• Creating additional features such as pickup hour, pickup day, and trip duration.

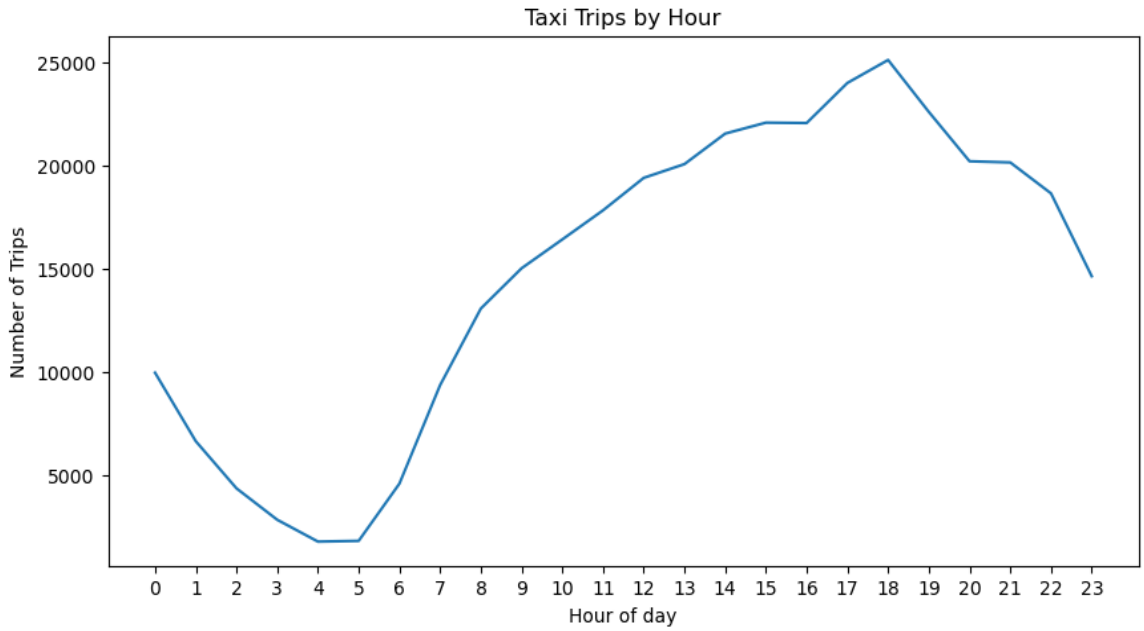
• Merging the taxi trip dataset with the taxi zone geographic dataset using location IDs.

After cleaning and preprocessing, the final dataset contained approximately 354,000 valid trip records used for further analysis.

Temporal Analysis: -

Hourly Trip Distribution

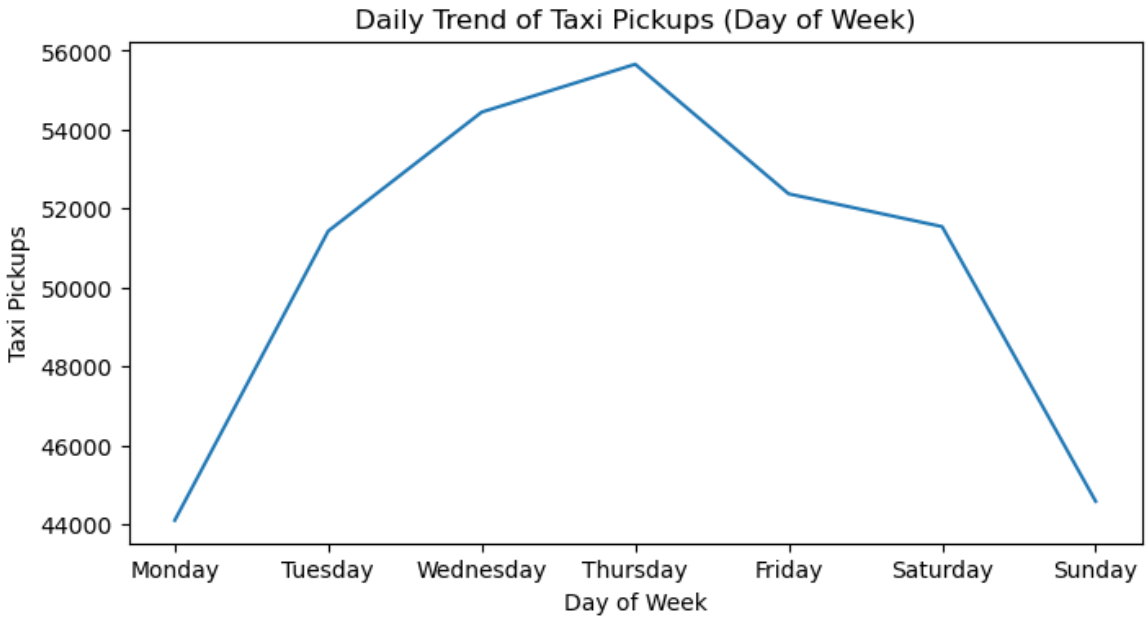
Taxi pickups were analysed across all 24 hours of the day to identify peak and off-peak periods.



Insight: Peak demand occurs between 5PM-7PM with over 24,000 trips per hour. The quietest period is 3AM-5AM with under 2,000 trips. Morning commute begins sharply at 6AM.

Daily Trip Distribution

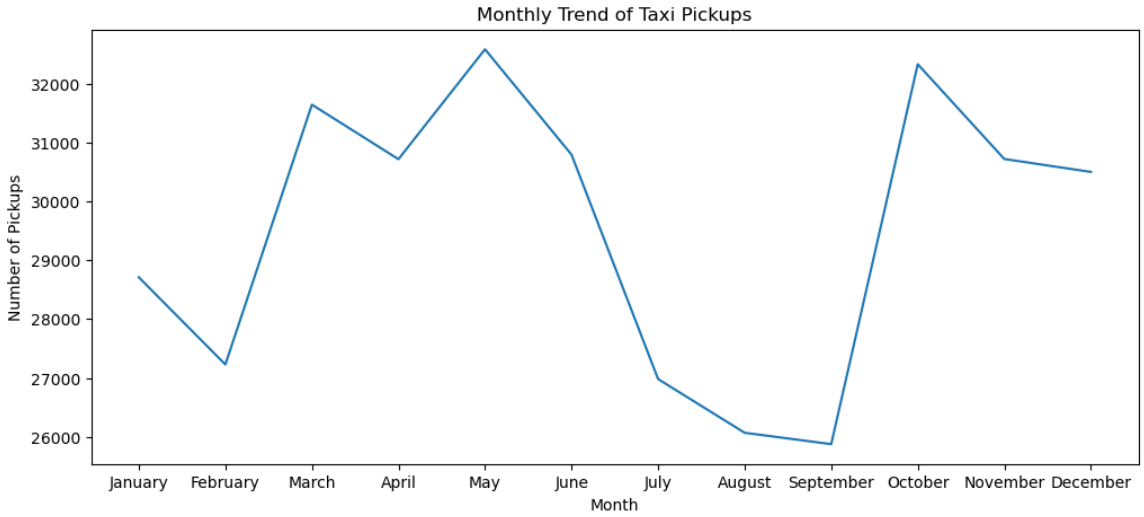
Taxi pickups were analysed across days of the week to identify busiest and quietest days.



Insight: Thursday is the busiest day with 55,660 trips. Monday and Sunday are the quietest days. Mid-week sees peak demand likely due to business activity.

Monthly Trip Distribution

Taxi pickups were analysed across all 12 months to identify seasonal patterns.

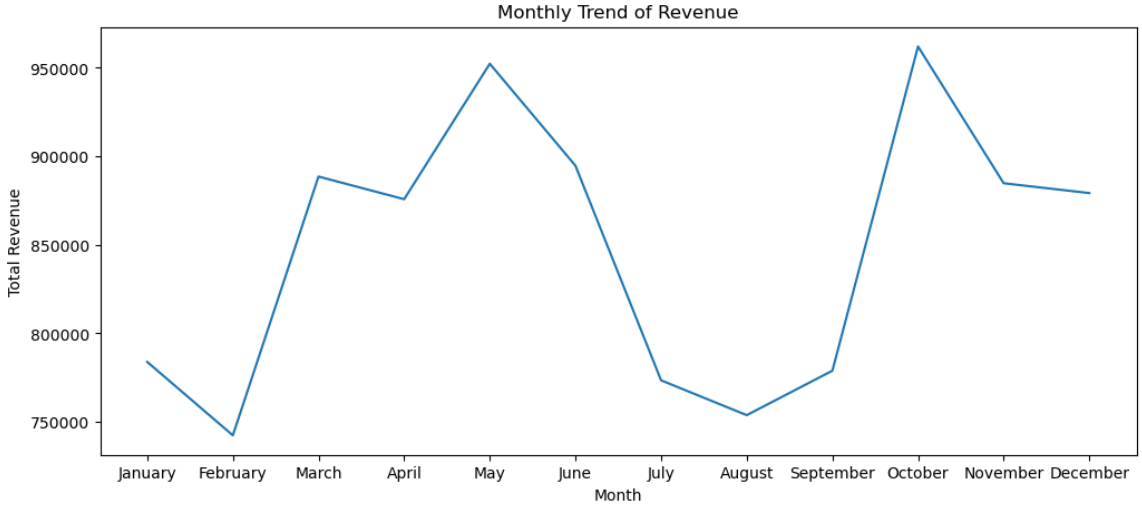


Insight: May and October are the busiest months with 32,580 and 32,326 trips respectively. July-September shows a dip likely due to summer vacations. Q2 and Q4 are the strongest quarters overall.

Financial Analysis: -

Monthly Revenue Trend

Total revenue was analysed across all 12 months to identify revenue patterns and seasonal trends.



Insight: May ($952K) and October ($962K) are the highest revenue months. February ($742K) is the lowest. Revenue closely follows trip volume patterns throughout the year.

Quarterly Revenue Share

Revenue was analysed by quarter to understand seasonal contribution to annual revenue.

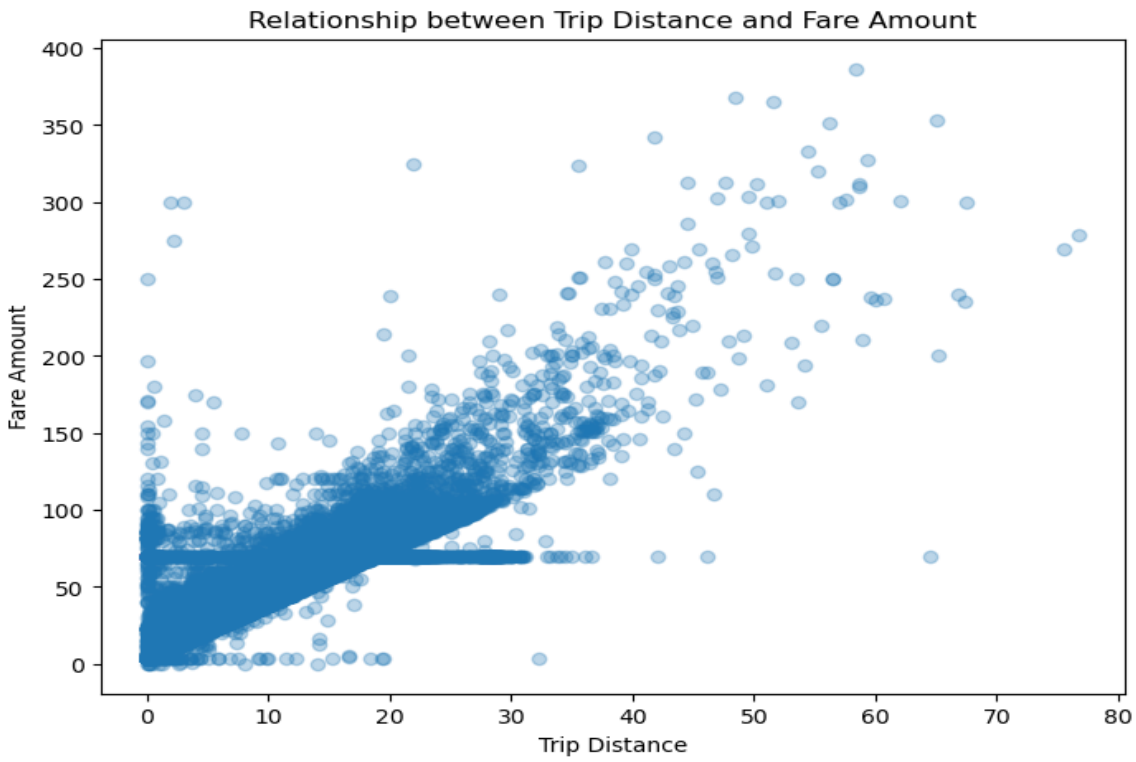
It is seen that the revenue for Q1 and Q4 exceeds that of Q2 and Q3 which comes around 2.72 million dollars. This is likely due to holiday seasons and tourist footfall.

Night vs Day Revenue Share

Insight: Daytime (6AM-10PM) contributes 87.91% of total revenue while nighttime (11PM-5AM) contributes 12.09%. However, night trips tend to be longer distance generating higher revenue per trip.

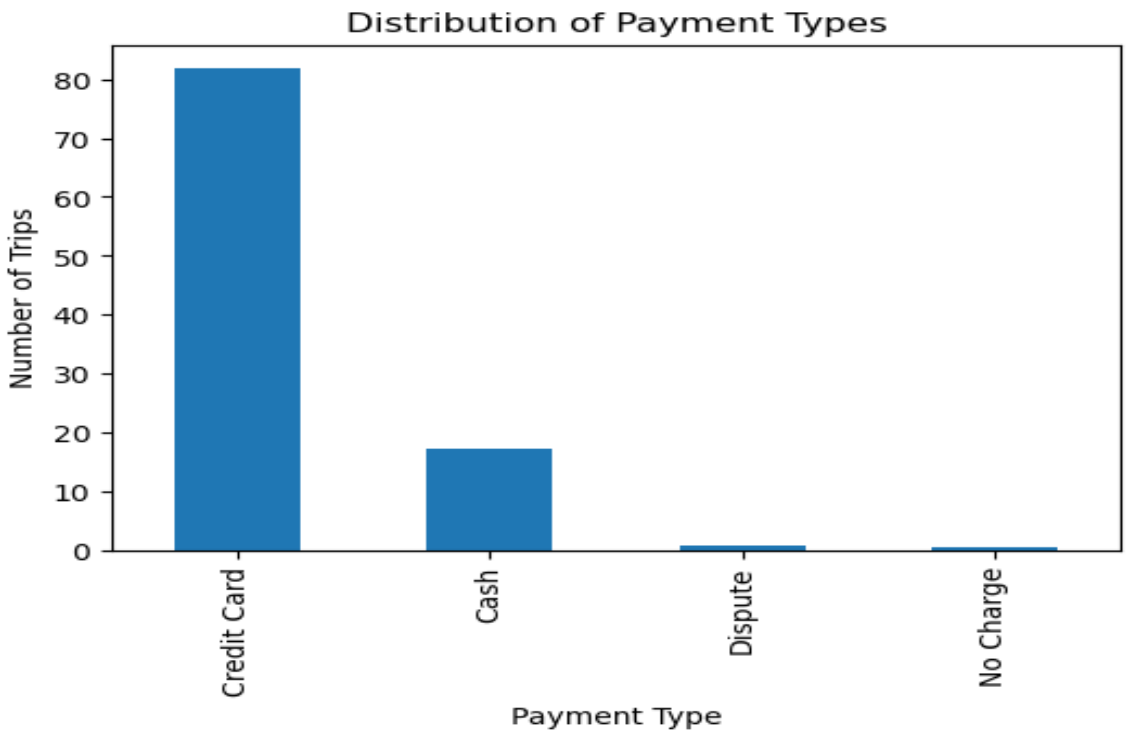
Trip Distance vs Fare Amount

Correlation between trip distance and fare amount: 0.947 (very strong positive correlation)



Insight: Trip distance is the strongest driver of fare amount with a correlation of 0.95. Longer trips cost significantly more as expected from the metered fare system.

Payment Type Distribution

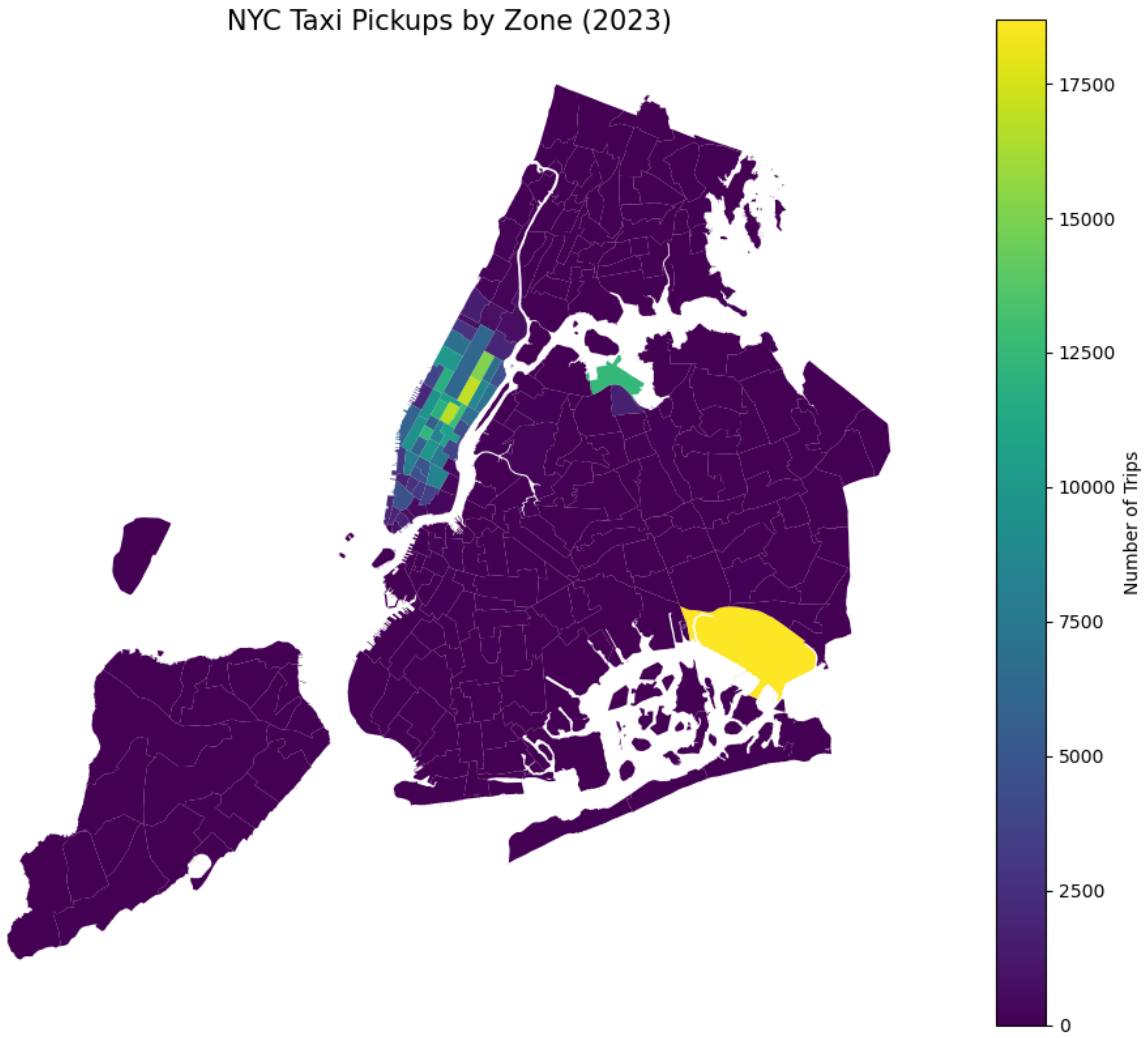


Insight: Over 81% of passengers pay by credit card while 17% pay cash. This is important for tip analysis as cash tips are not recorded in the system.

Geographical Analysis: -

Trip distribution was analysed across NYC taxi zones using GeoPandas and shapefile data.

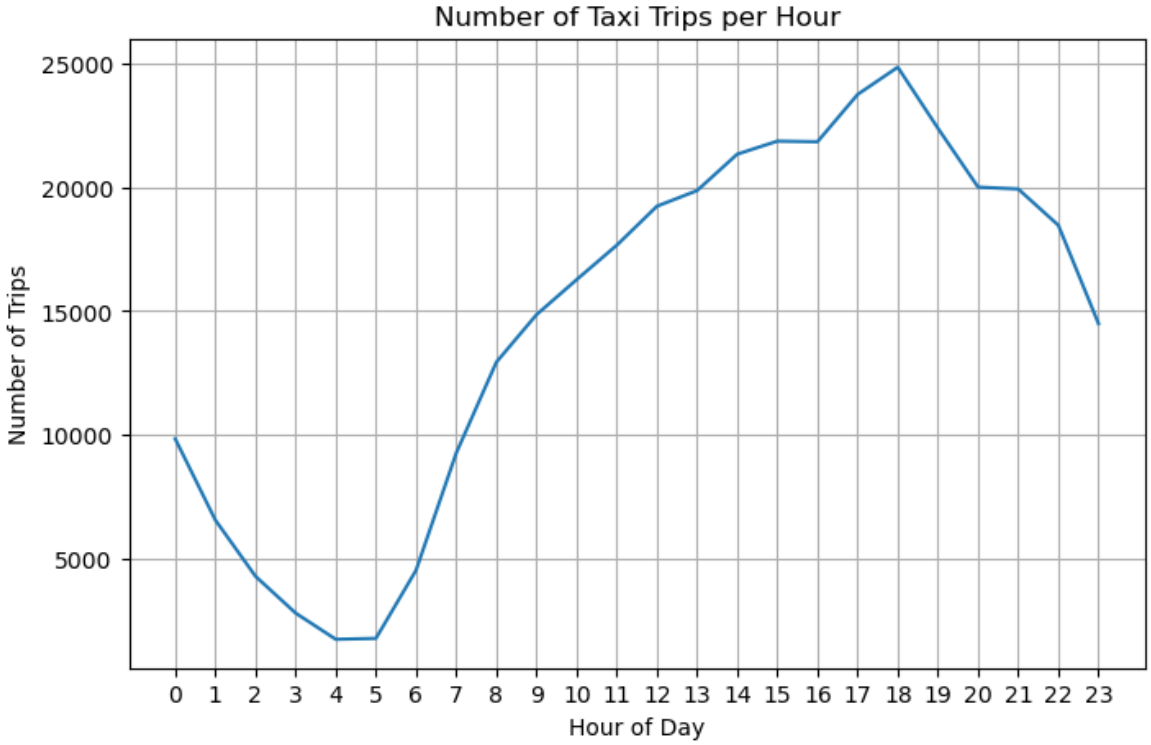
Zone-wise Trip Distribution



Insight: Manhattan dominates taxi pickups with 8 of the top 10 zones. JFK Airport leads all zones with 18,698 trips followed by Upper East Side South (16,986) and Midtown Center (16,741). Both major airports appear in the top 10 confirming high airport-related demand.

Operational Efficiency: -

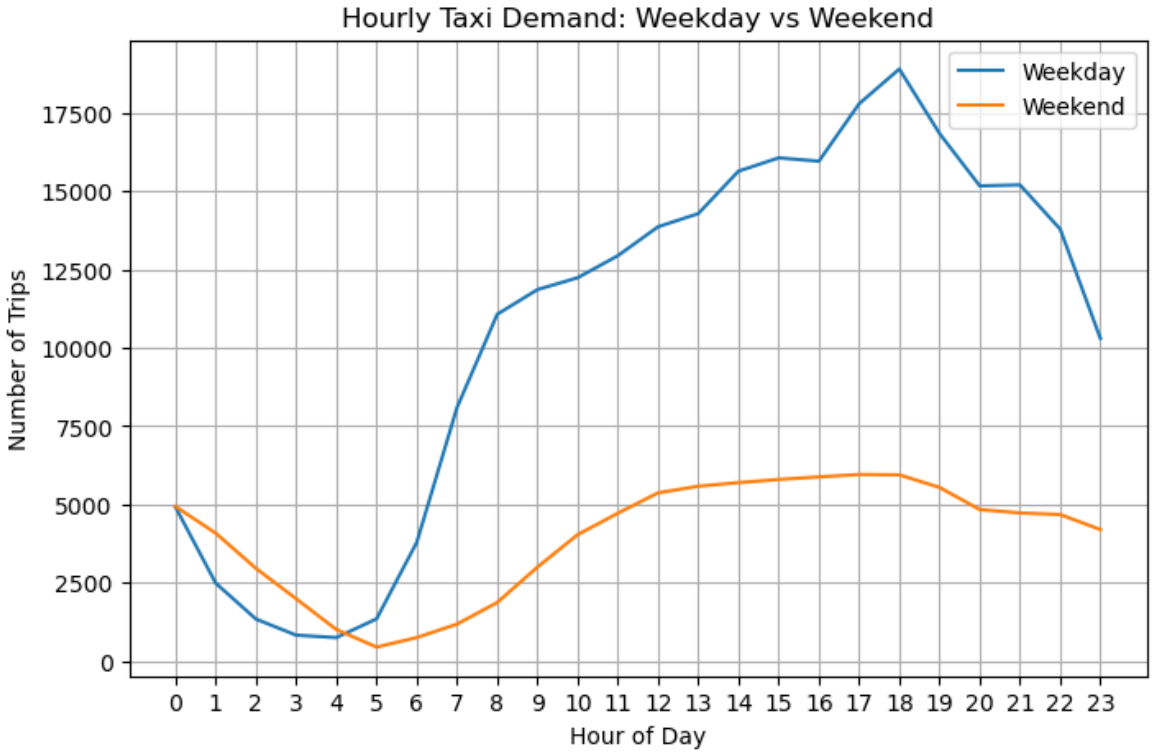
Busiest Hours and Scaled Trip Counts



The busiest hour is 6PM with 24,856 sampled trips. Scaling up by sampling fraction gives estimated actual trips:

Hour	Estimated Actual Trips
6PM (18:00)	2,485,600
5PM (17:00)	2,375,600
7PM (19:00)	2,238,900
3PM (15:00)	2,186,900
4PM (16:00)	2,184,400

Weekday vs Weekend Traffic



Insight: Weekdays show a classic double peak - morning rush (8AM-9AM) and evening rush (5PM-7PM). Weekends show a single gradual peak from noon to midnight reflecting leisure travel patterns.

Pickup/Dropoff Ratios by Zone

Zones with high pickup/drop-off ratios have more outbound demand than inbound -taxis leave full but return empty. Zones with low ratios have the opposite problem.

Zone	Pickups	Drop-offs	Ratios
East Elmhurst	1,626	176	9.24
JFK Airport	18,698	3737	5.00
LaGuardia Airport	12,475	4193	2.98
Penn Station	12,312	7955	1.55
Astoria	131	982	0.13

Insight: Airport zones have the highest ratios - taxis are always needed for outbound trips. Low ratio zones like Astoria have taxis arriving full but leaving empty indicating inefficient routing.

Pricing Strategy: -

Fare per Mile by Hour and Day

Average fare per mile was analysed across hours and days to identify natural pricing patterns.

Metric	Value
Highest fare/mile hour	4AM ($16.33/mile)
Lowest fare/mile hour	8-9PM (~$9.49/mile)
Highest fare/mile day	Thursday ($12.26/mile)
Lowest fare/mile day	Tuesday ($10.16/mile)

Insight: 4AM shows highest fare per mile due to long airport runs. Thursday and Sunday command higher rates naturally suggesting these are good candidates for surge pricing.

Fare per Mile per Passenger

As passenger count increases, fare per mile per passenger decreases - group rides are more cost efficient per person. Solo riders pay the most per mile while groups of 4 pay the least as the fare split among the passengers in group ride.

Vendor Fare Comparison by Distance Tier

Fare per mile was compared between Vendor 1 (Creative Mobile Technologies) and Vendor 2 (VeriFone) across three distance tiers.

Distance Tier	Trip Share	Creative Mobile ($/mile)	VeriFone ($/mile)
Short (0-2 miles)	55%	$9.79	$17.42
Medium (2-5 miles)	28%	$6.42	$6.55
Long (5+ miles)	17%	$4.45	$4.50

Insight: VeriFone charges nearly double for short trips which make up 55% of all NYC trips. For medium and long trips both vendors are nearly identical. Partnering with VeriFone would significantly boost revenue on the majority of trips.

Customer Experience: -

Tip Percentage Analysis

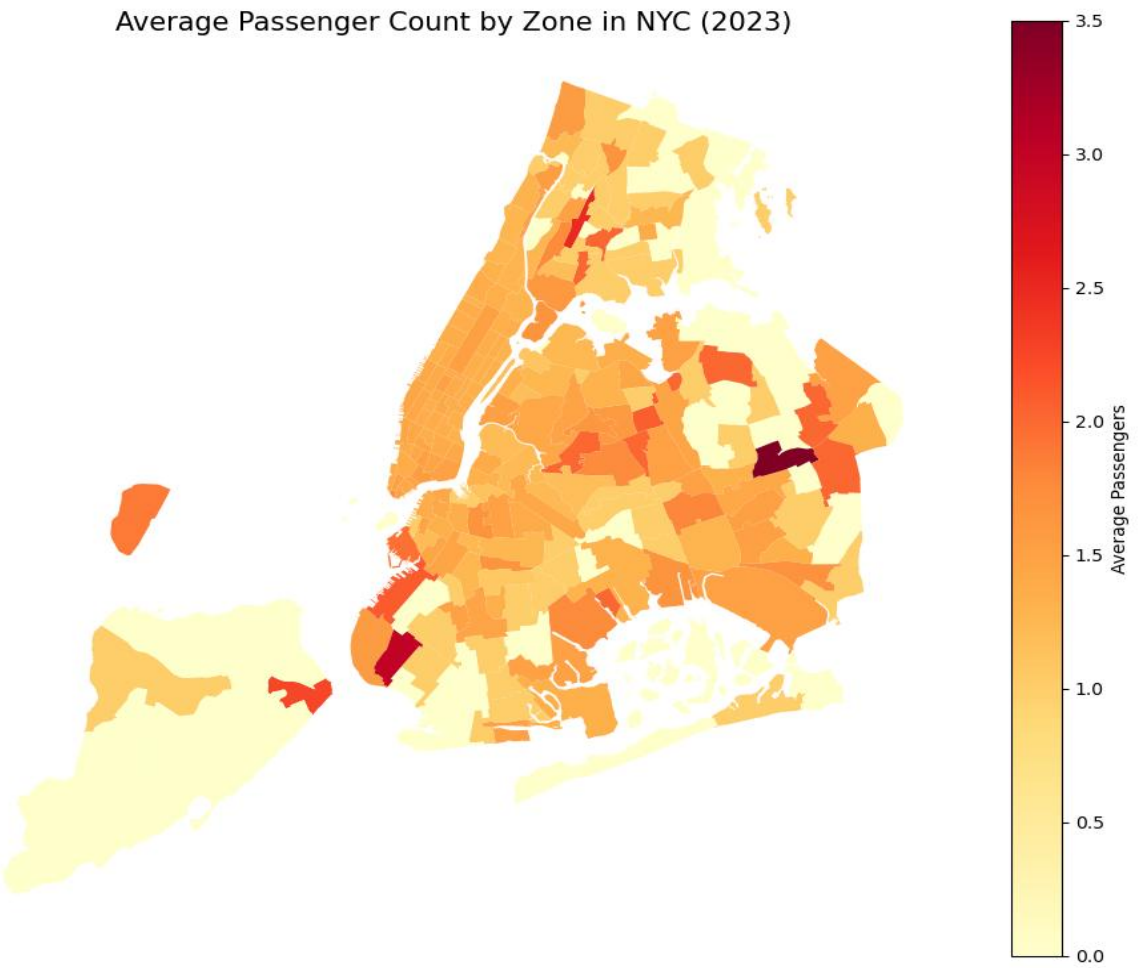
Tip percentages were analysed based on trip distance, passenger count and pickup hour to understand tipping behaviour.

Factor	Highest Tips	Lowest Tips
Distance	Short trips (22.6%)	Long trips (16.8%)
Passenger Count	Solo riders (20.9%)	Groups of 4 (17.4%)
Hour of Day	6PM-10PM (21-22%)	4AM-6AM (17-18%)

Insight: Short trips get higher tip percentages as passengers feel obligated to compensate for small fares. Evening hours see the highest tips. Groups of 4 tip the least due to diffusion of responsibility effect.

Passenger Count by Zone

Average passenger count was analysed across all NYC zones and visualised on a map.



Insight: Outer borough zones like Jamaica Estates (3.5), Dyker Heights (3.0) and Corona (2.08) have the highest average passengers due to family travel and limited subway access. Bronx zones are dominated by solo riders averaging 1.0 passenger per trip.

Conclusions and Recommendations: -

Routing and Dispatching

1. Dispatch maximum cabs during 5PM-10PM on weekdays - busiest period with 23,000-25,000+ trips per hour.

2. Reduce fleet during 3AM-5AM which has the lowest demand under 2,000 trips per hour.

3. Weekdays need double peak coverage (8-9AM and 5-7PM). Weekends need single coverage from noon to midnight.

4. After dropping passengers in low ratio zones like Astoria (ratio 0.13), immediately redirect cabs to high demand zones to avoid empty waiting.

Cab Positioning Strategy

1. Always maintain cab presence at JFK Airport, Upper East Side and Midtown - top 3 pickup zones.

2. During night hours (11PM-5AM) position cabs near East Village, West Village and Clinton East as these are top nightlife zones.

3. Deploy larger capacity vehicles in outer borough zones like Queens and Brooklyn where average passenger count is 2-3.5.

Pricing Strategy

1. Partner with VeriFone — charges nearly double for short trips ($17.42 vs $9.79/mile) which make up 55% of all trips.

2. Implement surge pricing during 4AM, Thursday and Sunday which naturally command higher fare per mile.

3. Introduce group ride discounts for 3-6 passengers to incentivize shared rides and increase vehicle utilization.

4. Maintain airport fee structure as airport zones generate the highest surcharge revenue.